

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08

Live sessions

90 min

Per session

04 weeks

Course duration

**Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

**Built for beginners**

No prior Python or machine learning required

**Live code-along on Civil Engineering datasets**

Materials · Buildings & Energy · Transport · Environment · Geotechnics

**Hands-on notebooks, cross-domain assignments & solutions**

Plus certificate & recordings (till 2 weeks after the course)

INSTRUCTOR**Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad



Research Excellence Award 2026, IIT Hyderabad



Invited talks at Oregon State University (USA) and a Faculty Development Programme at D.Y. Patil (Pune, India)



Resource person, five-day ML workshop at NIT Srinagar

Course outline

01 Python for Civil Engineers

AI/ML basics, Colab, NumPy & pandas with engineering data

02 From Raw Data to ML-Ready

Explore, clean, visualize & split engineering data

03 Your First Prediction Model

Linear regression, error metrics & cross-validation

04 Trees & Ensemble Models

Decision Trees, Random Forest & boosting for nonlinear problems

05 Classification for Engineering Decisions

Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks

Architecture, activations, training & overfitting

07 Explainable AI & Engineering Judgement

Feature importance, SHAP, and engineering interpretation

08 From Prediction to Probability

Monte Carlo, ML surrogates & probability of failure

Designed for Civil Engineers across disciplines — Materials, Structures, Geotechnics, Transportation, Water & Environment.

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty / Practising engineers / Complete beginners to AI & ML



**Saturday,
17 October 2026**

Scan to register · qudelastudios.com →

